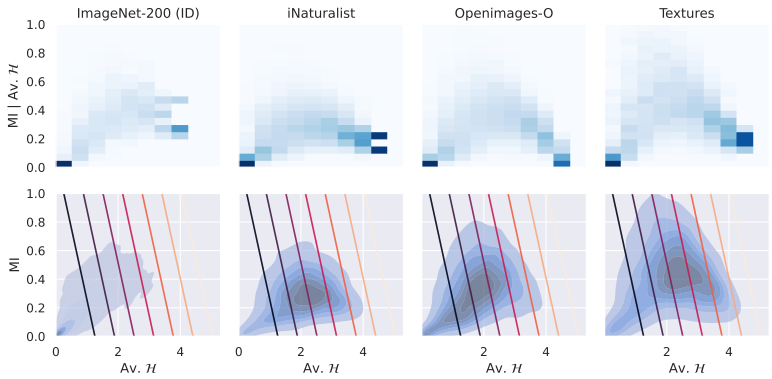

Figure 2. Top: empirical conditional histograms of MI|Av.$\mathcal{H}$. The behaviour of MI is quite similar between ID and OOD data, with the exception of Textures being slightly higher. Bottom: joint kernel density estimate plots of MI and Av.$\mathcal{H}$. Contours represent increasing (lighter) values of Ens.$\mathcal{H}$. MI is low for high Av.$\mathcal{H}$ OOD data. MI does not contribute significantly to Ens.$\mathcal{H}$.

Model	ID:	ImageNet-200 Method	OOD mean AUROC↑	OOD mean FPR@95↓	Near-ImageNet-200 AUROC↑	Near-ImageNet-200 FPR@95↓	Caltech-45 AUROC↑	Caltech-45 FPR@95↓	Openimage-O AUROC↑	Openimage-O FPR@95↓	iNaturalist AUROC↑	iNaturalist FPR@95↓
ResNet-50 ens. %Err: 16.28 av. %Err: 19.01	**Single**	MSP	84.47	66.55	77.56 ±1.2	82.21 ±1.1	84.54 ±1.2	67.83 ±4.0	85.09 ±1.0	71.43 ±1.8	88.40 ±0.9	60.83 ±2.9
		$\mathcal{H}$	85.98	62.12	78.18 ±1.2	82.33 ±1.7	86.13 ±1.0	63.96 ±3.6	86.21 ±1.0	70.76 ±2.0	89.64 ±0.8	59.22 ±3.5
		Energy	88.56	47.71	75.96 ±1.2	82.57 ±2.1	88.23 ±0.8	54.32 ±3.7	86.43 ±0.6	62.24 ±1.6	90.78 ±0.4	46.18 ±1.9
	Ens.	MSP	86.67	63.53	79.28	81.27	86.71	64.48	87.24	68.39	89.95	59.15
		Ens. $\mathcal{H}$	88.21	57.97	79.70	80.85	88.15	60.04	88.14	67.75	91.05	57.19
		Av. $\mathcal{H}$	88.07	57.48	**79.75**	**80.40**	88.22	59.31	88.29	66.50	91.67	52.31
		MI	83.46	78.84	77.94	84.18	83.75	79.36	83.46	78.40	81.97	90.05
		Av. Energy	**90.16**	**43.38**	77.60	80.47	**89.99**	**48.23**	**88.30**	**57.44**	**92.43**	**40.14**

Model	ID:	ImageNet-200 Method	Textures AUROC↑	Textures FPR@95↓	Colonoscopy AUROC↑	Colonoscopy FPR@95↓	Colorectal AUROC↑	Colorectal FPR@95↓	Noise AUROC↑	Noise FPR@95↓	ImageNet-O AUROC↑	ImageNet-O FPR@95↓
ResNet-50 ens. %Err: 16.28 av. %Err: 19.01	**Single**	MSP	86.95 ±1.4	60.49 ±4.0	90.67 ±1.8	53.16 ±9.0	89.82 ±5.2	50.88 ±11.9	81.70 ±6.5	71.13 ±15.6	75.50 ±0.6	80.96 ±0.7
		$\mathcal{H}$	88.69 ±1.3	53.84 ±3.8	92.94 ±1.6	42.88 ±9.1	91.95 ±4.9	41.10 ±15.7	83.61 ±5.3	64.78 ±22.8	76.51 ±0.7	80.18 ±0.3
		Energy	92.93 ±1.0	32.65 ±2.8	95.40 ±1.5	28.66 ±12.1	98.25 ±1.7	9.87 ±9.7	91.99 ±7.6	**38.43 ±23.3**	77.10 ±0.9	74.47 ±1.3
	Ens.	MSP	90.19	53.49	92.93	46.49	90.98	49.22	84.43	71.29	78.30	77.95
		Ens. $\mathcal{H}$	91.86	44.92	95.22	31.60	93.63	36.18	86.95	65.79	79.17	77.45
		Av. $\mathcal{H}$	91.23	48.00	95.26	31.65	93.97	34.46	85.67	65.75	78.56	78.95
		MI	88.43	61.90	89.11	76.07	82.14	88.66	84.92	77.39	**79.41**	73.55
		Av. Energy	**94.55**	**27.07**	**96.87**	**19.31**	**99.08**	**4.48**	**94.00**	41.13	78.64	**72.15**

Table 1. %AUROC↑ and %FPR@95↓ for different uncertainty scores over all OOD datasets. We show mean±2std. for single model results. **Bold** is best performance, underline 2nd or 3rd best. MI performs poorly compared to other uncertainties and even single models. Ens.$\mathcal{H}$ and Av.$\mathcal{H}$ consistently improve over single model $\mathcal{H}$. Av. Energy is able to improve over single model Energy and is best overall.

We can thus see how averaging $U(=\mathcal{H})$ over the ensemble leads to better OOD detection as the NLL of the ensemble OOD detector will be lower than the average NLL of the individual ensemble members.

The *key intuition* is that by ensembling binary classifiers that are diverse such that they make OOD detection errors on different data samples, the ensemble members are able to compensate for each other's miss-predictions and raise the ensemble performance above their individual average.

4.2. Using a Better OOD Detection Score

Existing work benchmarking Deep Ensembles focuses exclusively on uncertainty scores U extracted from the softmax outputs of the members. Given the explanation in the previous section, it is natural to ask whether we can improve performance by ensembling a better U. Using Energy [22] ($U = -\log \sum_k \exp v_k$, where $\boldsymbol{v}$ are the logits), and averaging over the ensemble, it can be seen in Tab. 1 that average ensemble Energy is able to improve over single model Energy, as well as produce the best OOD detection results in the whole comparison. We remark that although we have chosen Energy here, we expect any U with better performance than single model $\mathcal{H}$ to receive a similar uplift upon ensembling.

5. Concluding Remarks

In this work, we show that the existing intuition, that the diversity of a Deep Ensemble is a useful indicator for whether data is OOD, does not hold for ImageNet-scale CNNs. We instead suggest an alternative explanation for Deep Ensembles' superior OOD detection performance compared to single models: OOD detection is binary classification, and individual Deep Ensemble members produce diverse errors on this binary classification task, so combining them together yields better performance.

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